

Dear Vasiu,

Let $\mathcal{T}$ be a Tannakian category over an algebraically closed field k . Your question amounts to: does $\mathcal{T}$ always have a fiber functor over k ? I believe it is true. The following argument is presumably too pedestrian, hiding which “compactness arguments” are really relevant.

Let A be the set of strictly full subcategories of $\mathcal{T}$, stable by $\otimes$, subquotients and duals. Let $A_f \subset A$ be the set of those generated by finitely many objects, using the same operations. For $\alpha \in A$, I note $\mathcal{T}_\alpha$ the corresponding subcategory (as defined: $\mathcal{T}_\alpha = \alpha$). The set A is ordered by inclusion. I assume we already know the existence and uniqueness, up to isomorphism, of fiber functors for the $\mathcal{T}_\alpha$, $\alpha \in A_f$.

If $\mathcal{T}_\alpha \subset \mathcal{T}_\beta$, it makes sense to say that a fiber functor ω_β on $\mathcal{T}_\beta$ extends (or the nose) a fiber functor ω_α on $\mathcal{T}_\alpha$. If an extension, up to isomorphism, exists, an actual extension exists too.

Let us order the set of $(\alpha, \omega_\alpha) : \alpha \in A, \omega_\alpha$ fiber functor on $\mathcal{T}_\alpha$, by “ ω_β extends ω_α ”. To avoid set theoretical difficulties, one could consider only the fiber functors ω_α such that the $\omega_\alpha(X)$ take values in the set of vector spaces k^n , $n \in \mathbb{N}$.

The ordered set of (α, ω_α) is inductive (Bourbaki Ens Ch 3 §2.4): if I is a totally ordered subset, the “union” of the (α, ω_α) is a $\mathcal{T}_0$ in A ($= \bigcup_{(\alpha, \omega_\alpha) \in I} \mathcal{T}_\alpha$) with the fiber functor ω_0 characterized by $\omega_0|_{\mathcal{T}_\alpha} = \omega_\alpha$. This “union” majorizes I . By Bourbaki loc. cit. Th 2, the ordered set of the (α, ω_α) has a maximal element: $(\mathcal{T}_1, \omega_1)$. To prove that $\mathcal{T}_1 = \mathcal{T}$, it suffices to prove the following

Lemma 1. Let $\mathcal{T}'$ be in A and $\mathcal{T}''$ be in A_f . Let $\langle \mathcal{T}', \mathcal{T}'' \rangle$ be generated by $\mathcal{T}'$ and $\mathcal{T}''$; it is in A . Then, any fiber functor ω on $\mathcal{T}'$ can be extended (on the nose, or up to isomorphism, this amounts to the same) to $\langle \mathcal{T}', \mathcal{T}'' \rangle$.

I first prove

Lemma 2. Suppose $\mathcal{T}'$ is in A_f too. “Restriction” is then an equivalence of categories

$$(\text{fiber functors on } \langle \mathcal{T}', \mathcal{T}'' \rangle) \longrightarrow (\text{triples of})$$

a fiber functor ω' on $\mathcal{T}'$, ω'' on $\mathcal{T}''$ and an isomorphism τ of the restrictions of ω' and ω'' to $\mathcal{T}' \cap \mathcal{T}''$ (also in A_f).

We may assume that $\langle \mathcal{T}', \mathcal{T}'' \rangle$ is the category of representations of a group G , and that for invariant subgroups A and B , $\mathcal{T}'$ (resp. $\mathcal{T}''$) is the subcategory of representations where A (resp. B) acts trivially. That they generate $\langle \mathcal{T}', \mathcal{T}'' \rangle$ means that $A \cap B = \{e\}$. The intersection $\mathcal{T}' \cap \mathcal{T}''$ is the category of representations on which AB acts trivially.

The triples $(\omega', \omega'', \tau)$ are all isomorphic: as all ω' (resp. all ω'') are isomorphic, it suffices to see that $(\omega', \omega'', \tau_1)$ and $(\omega', \omega'', \tau_2)$ are isomorphic. Indeed τ_1 and τ_2 differ by an automorphism of $\omega'|_{\mathcal{T}' \cap \mathcal{T}''}$, and such an automorphism lifts to an automorphism of ω' :

$$G/A(k) \longrightarrow G/AB(k) \quad \text{is onto.}$$

We hence have here categories with just one isomorphism class of objects, and the question is to compare automorphism groups. We need to check

$$G \xrightarrow{\sim} \{(g', g'') \in G/A, G/B \mid g' \text{ and } g'' \text{ have same image in } G/AB\}$$

$$\begin{array}{ccc} G & \longrightarrow & G/A \\ \downarrow & & \downarrow \\ G/B & \longrightarrow & G/AB \end{array}$$

Injectivity of this morphism of groups amounts to $A \cap B = \{e\}$.

Surjectivity: if $\tilde{g}', \tilde{g}''$ lift g', g'' , $\tilde{g}'' = \tilde{g}' \cdot a \cdot b$ and $\tilde{g}'' \cdot b = \tilde{g}' \cdot a \in G$ maps to (g', g'') .

Lemma 3. Same as lemma 2, but $\mathcal{T}'$ only assumed to be in A .

Let B be the set of $\mathcal{T}_\alpha$ ($\alpha \in A_f$) contained in $\mathcal{T}'$. One has $\langle \mathcal{T}', \mathcal{T}'' \rangle = \bigcup_{\beta \in B} \langle \mathcal{T}_\beta, \mathcal{T}'' \rangle$. One has equivalences

$$(\text{fiber functors on } \mathcal{T}) \xrightarrow{\sim} (\text{fiber functors on the } \mathcal{T}_\beta,$$

plus a compatible system of isomorphisms $\omega_\beta|_{\mathcal{T}_\gamma} \xrightarrow{\sim} \omega_\gamma$ for $\mathcal{T}_\gamma \subset \mathcal{T}_\beta$; compatible: condition for $\mathcal{T}_\delta \subset \mathcal{T}_\gamma \subset \mathcal{T}_\beta$).

Same for fiber functors on the $\langle \mathcal{T}', \mathcal{T}'' \rangle = \bigcup \langle \mathcal{T}_\beta, \mathcal{T}'' \rangle$.

The $\mathcal{T}_\beta \cap \mathcal{T}''$ have a largest element: they if $\mathcal{T}'' = \text{Rep}(G'')$, they correspond to invariant subgroups of G'' , subgroups are closed subschemes, and one uses the noetherian property. If β_0 is such that $\mathcal{T}_{\beta_0} \cap \mathcal{T}''$ is the largest $\mathcal{T}_\beta \cap \mathcal{T}''$, for any $\beta > \beta_0$, extending $\omega_\beta = \omega|_{\mathcal{T}_\beta}$ to $\langle \mathcal{T}_\beta, \mathcal{T}'' \rangle$ amounts to extending $\omega_\beta|_{\mathcal{T}_{\beta_0} \cap \mathcal{T}''}$ to $\mathcal{T}''$ (lemma 2), that is to extend ω_{β_0} from $\mathcal{T}_{\beta_0} \cap \mathcal{T}''$ to $\mathcal{T}''$. If we choose one such extension, we get up to unique isomorphism a system of extensions of the ω_β to $\langle \mathcal{T}_\beta, \mathcal{T}'' \rangle$, and by gluing them an extension of ω to $\langle \mathcal{T}', \mathcal{T}'' \rangle$.

This is all we need to conclude that $\mathcal{T}' = \mathcal{T}$.

This proof should be cleaned up. After all, we are proving that some projective system of gerbs (of the fiber functors on the $\mathcal{T}_\alpha$, $\alpha \in A_f$), where “projective system” is taken in a 2-categorical sense, has a non empty projective limit (again limit in a 2-categorical sense). Maybe such a translation would make the “compactness arguments” used cleaner. They were two of them: “fiber functor is a property of finite type (cf Bourbaki Ens Ch 3 §4.5)”, and the noetherian property for subgroups.

The same argument gives uniqueness up to isomorphism of ω , over k ; we get a maximal $\mathcal{T}_\alpha$ ($\alpha \in A$) over which we have an isomorphism, and extend further if $\mathcal{T}_\alpha \neq \mathcal{T}$.

Best,
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